SKILL 5: LOGISTIC REGRESSION CLASSIFICATION

One Piece Bounty Project

Aim. Explore bounty records, engineer name-based features, and review Logistic Regression classification of high-bounty characters.

Dataset summary. The supplied report records 165 raw entries, 144 usable Berry records, and a 332,500,000-Berry classification threshold. The report's class counts are 108 standard and 36 high bounty.

Method. The reported pipeline is StandardScaler followed by LogisticRegression. Name-derived variables include length, word count, uppercase count, vowel measures, initials, D. marker, title marker, and longest-word length.

Metric	Reported value
Accuracy	0.7241
Precision	0.4286
Recall	0.4286
F1 score	0.4286
ROC AUC	0.6364
Average precision	0.6082
Log loss	0.5372
Best F1 threshold	0.76
Out-of-fold ROC AUC	0.6844

Interpretation. The reported test accuracy is 0.7241 and ROC AUC is 0.6364. Precision, recall, and F1 are each 0.4286. These values indicate moderate separation and suggest that the high-bounty class is not consistently identified. The class imbalance makes recall and precision important alongside accuracy.

Limitations. Name-based correlations are exploratory and do not imply causation. The uploaded raw CSV includes currency-formatted values and possibly unknown formats; recreated charts may therefore have counts that differ from the report's documented processed subset. The reported metrics above are preserved as supplied.

Figure 1. Bounty distribution from the uploaded CSV.

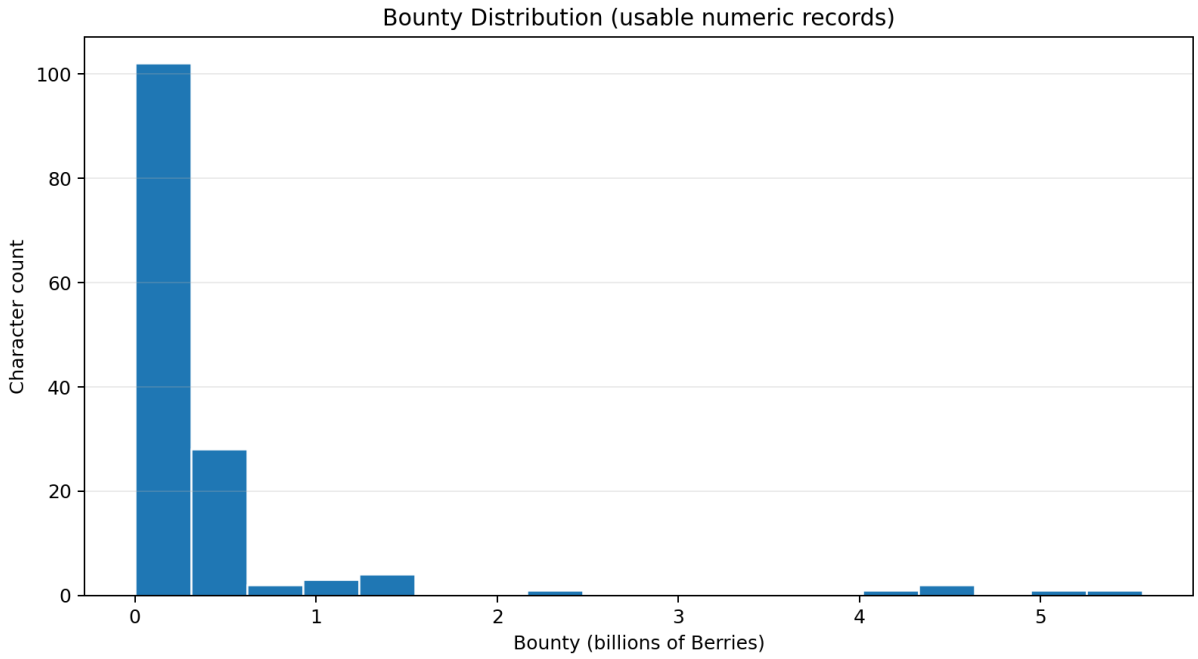


Figure 2. Recreated class counts using the documented threshold.

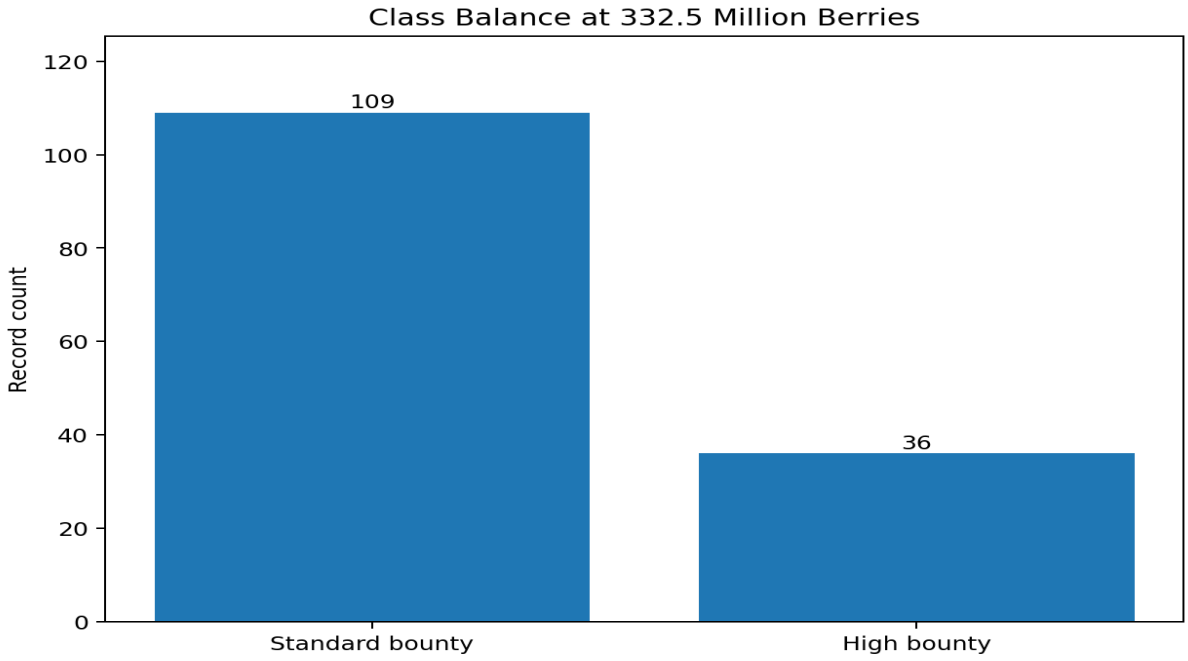


Figure 3. Fifteen largest numeric bounties.

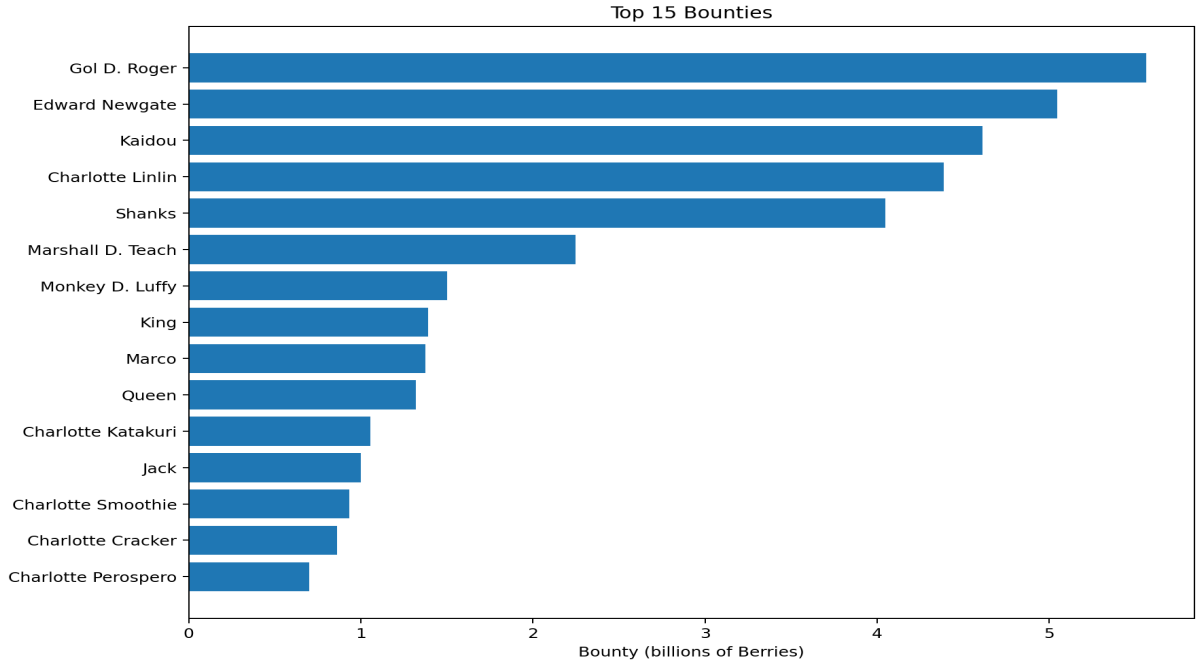


Figure 4. Correlation of engineered name features with the high-bounty label.

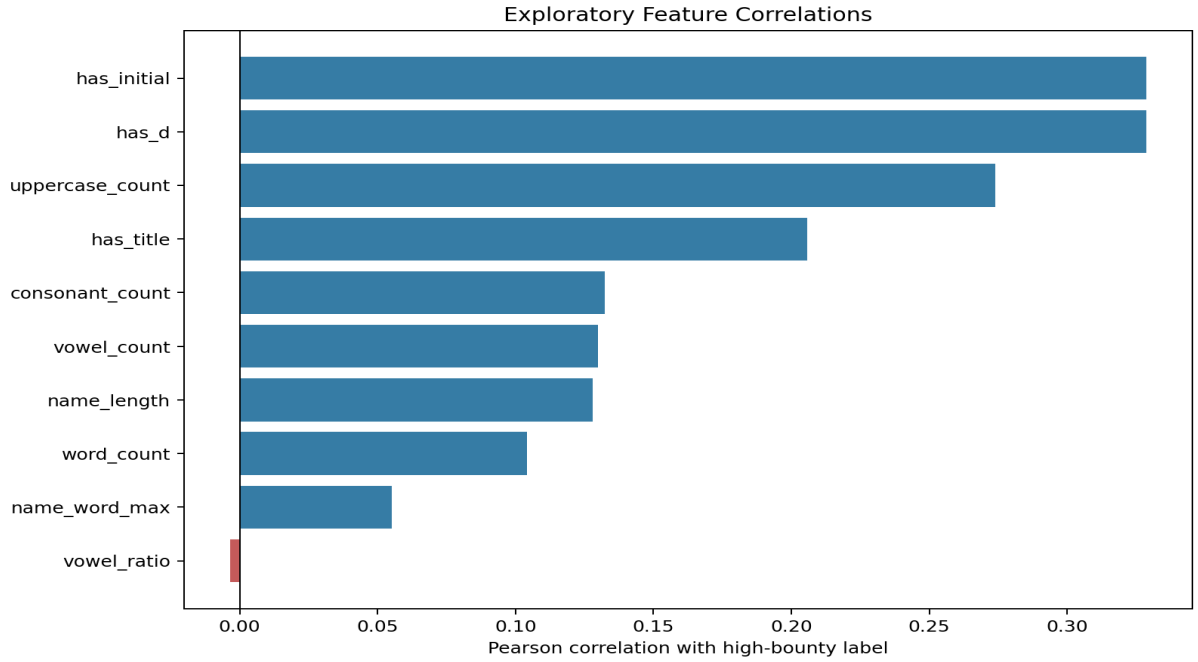


Figure 5. Name length grouped by bounty class.

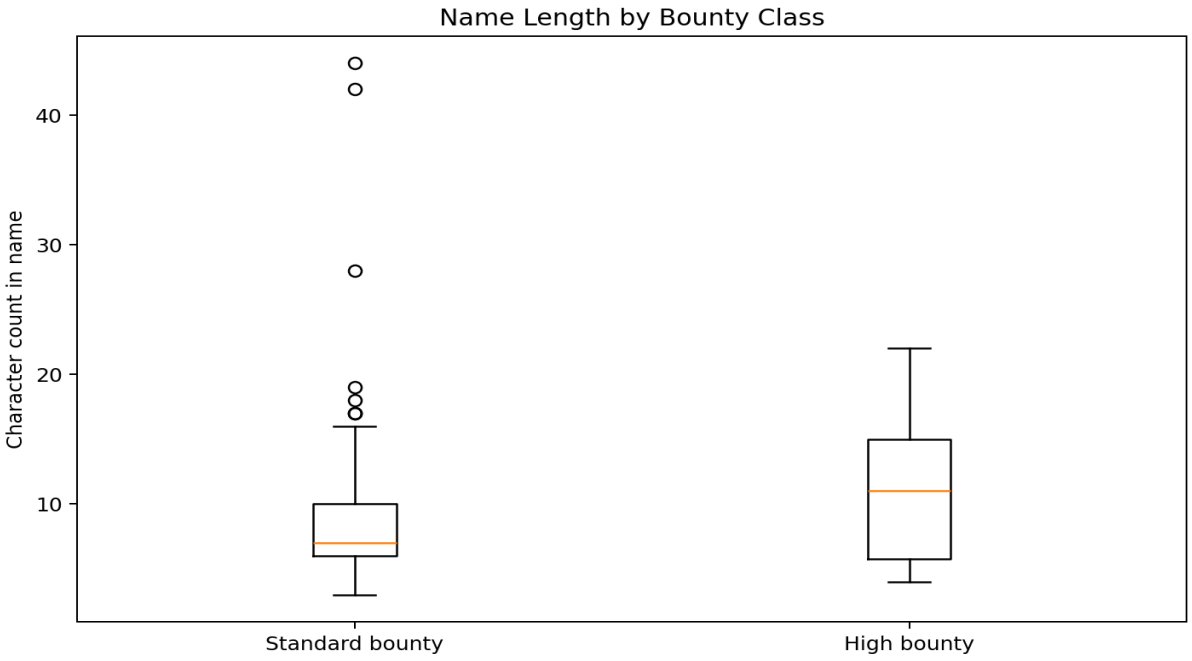


Figure 6. Metrics reproduced from the supplied report.

